Key

Style 5

- Truth and Nil 6
- Arithmetic 16
- Constants and Chance 38
- Number Theory 49
- Polynomials and Roots 56
- Matrices 63
- Arrays 71
- Boxes 86
- Sets and Membership 97
- Trains 106
- Functions 124
- Errors 133
- Classes and Properties 144
- The Virtual 154

Dungeon 160

- Places 161
- Classes 165
- Objects 171
- Entities 176
- Food 179
- Species inheritance 198

Index 240

- English Index 241
- Glyph Index 248

Contents

Style 5

Truth and Nil 6

Niladic 6

Monadic 8

Dyadic 10

Arithmetic 16

Niladic 16

Monadic 20

Dyadic 31

Constants and Chance 38

Niladic 38

Monadic 43

Dyadic 46

Number Theory 49

Monadic 49

Dyadic 52

Polynomials and Roots 56

Monadic 56

Dyadic 59

Matrices 63

Niladic 63
Monadic 65
Dyadic 68
Arrays 71
Niladic 71
Monadic 74
Dyadic 81
Boxes 86
Niladic 86
Monadic 88
Dyadic 92
Sets and Membership 97
Niladic 97
Monadic 99
Dyadic 102
Trains 106
Niladic 106
Monadic 109
Dyadic 121
Functions 124
Monadic 124
Dyadic 129
Errors 133
Niladic 133
Monadic 139
Classes and Properties 144
Niladic 144
Monadic 147

Dyadic 150
The Virtual 154
Niladic 154
Monadic 157
Dungeon 160
Places 161
Classes 165
Objects 171
Entities 176
Food 179
Species inheritance 198
Index 240
English Index 241
Glyph Index 248

Style

Truth and Nil: Niladic

Wuxia

俠

The Hero

U+4FA0

Nil



The empty set — quiet failure value; absorbs; Nil⊡function → Nil

U+2205

Truth

⊤

Verum — first-class truth; no true anymore

U+22A4

Truth and Nil: Monadic

Invert

~

Invert — monadic only (tilde). Scalar: ~ 0 → 1, ~ 1 → 0 (user wording to confirm). Unary scalar verbs operate on array back ranks (new 2026-10-11)

U+007E

Not

¬

Not — monadic boolean not: ¬ Truth yields Nil, ¬ Nil yields Truth (new 2026-10-14)

U+00AC

Congruent

≡

Ambivalent: monadic binds train to lhs⊡function

U+2261

Truth and Nil: Dyadic

Less Than



Less Than — dyadic < A B yields Truth when A < B, Nil otherwise (new 2026-10-14)

U+003C

Greater Than



Greater Than — dyadic > A B yields Truth when A > B, Nil otherwise (new 2026-10-14)

U+003E

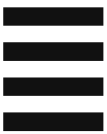
Congruent



Ambivalent: dyadic ≡ A B yields Truth when A and B have the same binding (same location in the system tables), Nil otherwise

U+2261

IsAt



IsAt — dyadic ≣ X Y yields Truth when X and Y share the same pointer address, Nil otherwise (new 2026-10-14)

U+2263

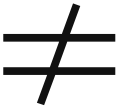
Equality



Equality — dyadic = A B yields Truth when A = B, Nil otherwise (new 2026-10-14)

U+003D

Unequal



Unequal — dyadic ≠ A B: sugar for ¬ = A B (boolean Not of Equality) (new 2026-10-14)

U+2260

Less Or Equal



Less Or Equal — dyadic ≤ A B yields Truth when A ≤ B, Nil otherwise (new 2026-10-14)

U+2264

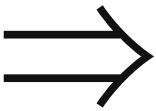
Greater Or Equal



Greater Or Equal — dyadic ≥ A B yields Truth when A ≥ B, Nil otherwise (new 2026-10-14)

U+2265

Implies



Implies — logical implication; ∅ ⇒ ∅ → ⊤

U+21D2

forall



Train quantifier — every element must evaluate non-nil. Idiom: dyadic NAND is ¬ ∀ 𝕀𝕀 X Y

U+2200

Approx



Approx — dyadic ≈ A B yields Truth when|A - B|is within 1e-6, Nil otherwise (new 2026-10-14)

U+2248

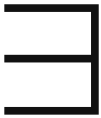
UnApprox



UnApprox — dyadic ≉ A B: sugar for ¬ ≈ A B (boolean Not of Approx) (new 2026-10-14)

U+2249

Forsome



Existential — at least one element evaluates non-nil

U+2203

Fornone



Not-forsome — defined as ~Forsome

U+2204

Arithmetic: Niladic

Signum / Plus

+

Ambivalent: Niladic + yields 0.
(Was ∓ U+2213, retired 2026-10-12

U+002B

Negate / Subtract

−

Niladic - yields 0 (new 2026-10-11

U+002D

Plus (Complex)

⨢

Plus (Complex), aka Sum Complex, Add Complex —; Niladic: sugar for ℿ 0 0, the complex zero (new 2026-10-14)

U+2A22

Times (Complex)

⨣

Times (Complex) —; Niladic: sugar for ℿ 1 0, the complex one (new 2026-10-14)

U+2A30

Minus (Complex)



Minus (Complex) —; Niladic: sugar for ⨿ 0 0, the complex zero (new 2026-10-14)

U+235B

Times (Polar)



Times (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)

U+2A02

Divide (Complex)



Divide (Complex) —; Niladic: sugar for ⨿ 1 0, the complex one (new 2026-10-14)

U+2A2C

Divide (Polar)



Divide (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)

U+2A38

Factorial

!

Factorial —; Niladic ! yields 1 (new 2026-10-12)

U+0021

Percent

%

Percent —; Niladic % is 0 (new 2026-10-12

U+0025

Square



Square — ⊠ X is X squared.; Niladic ⊠ yields 1, derived from Multiply (new 2026-10-12)

U+22A0

Multiply



*Multiply —; Niladic ⨯ yields empty array. Replaces Star * U+002A (new 2026-10-12; boxed forms 2026-10-12*

U+2A2F

Arithmetic: Monadic

Signum

+

Ambivalent: monadic Signum (less than zero yields −1, greater than zero yields 1, zero yields 0; back-rank scalar unary)

U+002B

Negate

-

Negate (monadic - X, unary minus)

U+002D

Conjugate

⊶

Conjugate — monadic ⊶X: complex conjugate; negates the imaginary part, preserving orientation (on [r,i] yields [r,−i]; on [r;i] yields [r;−i]) (new 2026-10-15)

U+22B6

Double

+2

Double — monadic +⍨X yields 2 × X (new 2026-10-15)

U+2A27

Sum

∑

Sum — sugar for reduce-addition: ∑ X means / + X (redefined 2026-10-12)

U+2211

Square Root

√

Square Root — monadic √ X yields the square root of X (new 2026-10-15)

U+221A

Plus (Complex)

⨁

Plus (Complex), aka Sum Complex, Add Complex —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.

U+2A22

Times (Complex)

⨂

Times (Complex) —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.

U+2A30

Minus (Complex)



Minus (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.

U+235B

Times (Polar)



Times (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.

U+2A02

Divide (Complex)



Divide (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.

U+2A2C

Divide (Polar)



Divide (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.

U+2A38

Percent



Percent —; monadic % X is absolute value, a back-rank operator.; monadic redefined 2026-10-12)

U+0025

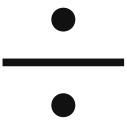
Star



*Star — exponentiate: monadic * X is 1, because the missing exponent Y is assumed 0, giving X^0 (new 2026-10-12). Idiom: X-th root of Y is * Y ÷ 1 X*

U+002A

Quotient



Quotient —; Monadic ÷ X: if X is a square array, yields the matrix inverse; else reciprocal, a back-rank scalar operator (new 2026-10-12)

U+00F7

Multiply



Multiply —; Monadic × X yields X when X numeric; on boxed X, catenates once.; monadic redefined 2026-10-12)

U+2A2F

Ceiling



Ceiling —; monadic ⌈X rounds up
(from APL; new 2026-10-12)

U+2308

Floor



Floor —; monadic ⌊X rounds down
(from APL; new 2026-10-12)

U+230A

Sine



Sine (new 2026-10-12)

U+25DD

Cosine



Cosine (new 2026-10-12)

U+25DE

Arccosine



Arccosine (new 2026-10-12)

U+25F6

Arcsine



Arcsine (new 2026-10-12)

U+25F7

Tangent



Tangent (new 2026-10-12)

U+2AD7

Arctangent



Arctangent (new 2026-10-12)

U+2AD8

Hyperbolic Sine



Hyperbolic Sine (new 2026-10-12)

U+2AA6

Hyperbolic Cosine



Hyperbolic Cosine (new 2026-10-12)

U+2AA7

Hyperbolic Arcsine



Hyperbolic Arcsine (new 2026-10-12)

U+2ABD

Hyperbolic Tangent



Hyperbolic Tangent (new 2026-10-12)

U+2AD1

Natural Log



Natural Log —; monadic ⍁ X is ln X.; Monadic ⍁ on a polar 2×1 yields the rectangular complex pair [ln r, θ].

U+237C

Antilog



Antilog — sugar for Star: ⍋̇ X is e raised to the X. Monadic ⍋̇ on a rectangular complex pair yields the polar 2×1 [e^a; b] (new 2026-10-12; glyph was 𝒆 U+1D58A until 2026-10-14)

U+29CA

Hyperbolic Arctangent



Hyperbolic Arctangent (new 2026-10-12)

U+2A4D

Hyperbolic Arccosine



Hyperbolic Arccosine (new 2026-10-12)

U+2ABE

Factorial

!

Factorial —; monadic ! X is X factorial

U+0021

Fourier



Fourier — monadic F X: FFT of X (new 2026-10-15)

U+03DC

InverseFourier



InverseFourier — monadic ⅁ X: inverse FFT of X (new 2026-10-15)

U+2132

Base Ten Log



Base Ten Log — ⍙ X is log base 10 of X, sugar for ⍦ 10 X (new 2026-10-12)

U+2A5F

Magnitude



Magnitude — monadic ⍫X: absolute value of X (new 2026-10-16)

U+2357

Arithmetic: Dyadic

Plus



Ambivalent: dyadic + X Y yields X + Y (addition).; dyadic 2026-10-15)

U+002B

Subtract

Subtract (dyadic - X Y yields X - Y, first minus second), the subtract that works with the / right-fold Reduce.; dyadic 2026-10-12)

U+002D

Plus (Complex)



Plus (Complex), aka Sum Complex, Add Complex —; dyadic: sum of complex numbers (1×2 rectangular), per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before adding.

U+2A22

Times (Complex)



Times (Complex) —; dyadic: product of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before multiplying.

U+2A30

Minus (Complex)



Minus (Complex) —; dyadic: difference of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before subtracting.

U+235B

Times (Polar)



Times (Polar) —; dyadic: product of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes multiply, angles add; if either or both X and Y are scalar, promote to polar 2×1 before multiplying.

U+2A02

Divide (Complex)



Divide (Complex) —; dyadic: quotient of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before dividing.

U+2A2C

Divide (Polar)



Divide (Polar) —; dyadic: quotient of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes divide, angles subtract; if either or both X and Y are scalar, promote to polar 2×1 before dividing.

U+2A38

Percent



Percent —; dyadic % X Y yields the remainder of X divided by Y

U+0025

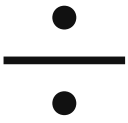
Star



*Star — exponentiate: dyadic * X Y is X to the power Y (left=base, right=exponent)*

U+002A

Quotient



Quotient —; dyadic ÷ X Y divides X by Y, yielding a real; division by zero yields infinity, except 0/0 which yields Nil.

U+00F7

Multiply



Multiply —; dyadic multiplies its arguments; when X is boxed, ⍪X Y catenates X repeated Y times.

U+2A2F

Factorial

!

Factorial —; dyadic ! X Y is the binomial coefficient Y choose X.

U+0021

Ceiling

⌈

Ceiling —; dyadic ⌈ X Y is maximum

U+2308

Floor

⌊

Floor —; dyadic ⌊ X Y is minimum

U+230A

Natural Log



Natural Log —; Dyadic ⍟ X Y is log base X of Y, sugar for ÷ ⍟ Y ⍟ X (new 2026-10-12)

U+237C

Square



Square — ⊠ X is X squared.; Dyadic ⊠ X Y squares X Y times.

U+22A0

Constants and Chance: Niladic

Half

½

Half —; niladic: 0

U+00BD

Pi

π

Pi — niladic π: value of pi (new 2026-10-14)

U+03C0

Bell

𝔅

Bell —; niladic: random point in (-1, 1), exclusive

U+1D56D

Real

ℜ

Real — niladic: random real number between 0 and 1; sugar for ½ + 1 𝔅 (new 2026-10-13)

U+1D57D

Gamma



Gamma — niladic γ: Euler-Mascheroni constant (new 2026-10-14)

U+03B3

Golden



Golden — niladic φ: golden mean (new 2026-10-14)

U+03C6

Euler



Euler — niladic ℇ: Euler's constant (new 2026-10-14)

U+2107

Dice



Dice —; niladic: single 6-sided die (1-6)

U+1D56F

Dice Percentile

‰

*Dice Percentile — niladic, intrinsic: A gets D10, B gets D10, yields + B * 10 A; the per mille sign U+2030 (2026-10-20)*

U+2030

D2



D2 — niladic: a single 2-sided die (1-2); sugar for 𝔇2; the die face U+2681 (2026-10-20)

U+2681

D4



D4 — niladic: a single 4-sided die (1-4); sugar for 𝔇4; the die face U+2683 (2026-10-20)

U+2683

D6



D6 — niladic: a single 6-sided die (1-6); sugar for 𝔇6; the die face U+2685 (2026-10-20)

U+2685

D10



D10 — niladic: a single 10-sided die (1-10); sugar for ↻10; the circled ten U+2469 (2026-10-20)

U+2469

Pick



Pick —; niladic: Nil

U+2A03

Constants and Chance: Monadic

Half

½

Half —; monadic ½ X: X divided by 2; array → sugar for ⊞(⍴ X); train → floor(length/2) index; dictionary → Nil (new 2026-10-13)

U+00BD

Pick

⍤

Pick —; monadic ⍤X: scalar → same scalar; train → use Dice of size train to select element; array → use Dice of sides from shape to select element; dictionary → random {key:value} pair (new 2026-10-13)

U+2A03

Bell

𝔅

Bell —; monadic 𝔅μ: bell curve point with mean μ (μ in (-1,1))

U+1D56D

Dice

𝔇

Dice —; monadic 𝔇X: single X-sided die (1-X)

U+1D56F

Deal



Deal — given a scalar, works as expected; given an array, fills the entire back rank with limited values. Concrete examples and summary TBD (new 2026-10-11)

U+003F

Constants and Chance: Dyadic

Bell

𝔅

Bell —; dyadic 𝔅 μ σ: bell curve point with mean μ and stddev σ (σ ≤ 1/3) (new 2026-10-13)

U+1D56D

Dice

𝔇

Dice —; dyadic 𝔇 X N: N X-sided dice; all sugar via ℜ (hence 𝔅) (new 2026-10-13; die face U+2685 tried and reverted 2026-10-20)

U+1D56F

Number Theory: Monadic

Divisible



Divisible —; monadic ⍱ X yields Truth when X is divisible by 2, Nil otherwise

U+23A2

Indivisible



Indivisible —; monadic ⍲ X: sugar for ¬ ⍱ X (the Odd test)

U+2AEE

Primes



Primes — monadic 𝒫X yields the prime factors of X, per J's q:. (new 2026-10-18)

U+1D4AB

Prime



Prime — monadic 𝔓 N yields the Nth prime, per J's p: (zero-indexed: 𝔓 0 is 2). (new 2026-10-18)

U+1D513

Prime



Prime — monadic ⌽ X: sugar for ⍵⍵/ X ⍳2 ⌊√ X ⍸ ≤ X 1 (new 2026-10-15)

U+23C0

Factor



Factor — factorization verb (new 2026-10-16)

U+27D2

Combinatoric



Combinatoric —; monadic ## X: factorial of X

U+2A69

Number Theory: Dyadic

LCM



LCM — dyadic ⍜ X Y: least common multiple of X and Y (new 2026-10-16)

U+238A

Divisible



Divisible —; dyadic ⍱ X Y yields Truth when X is divisible by Y, Nil otherwise (new 2026-10-15)

U+23A2

GCD



GCD — dyadic ⫻ X Y: greatest common divisor of X and Y (new 2026-10-16)

U+2980

Indivisible



Indivisible —; dyadic ⍲ X Y: sugar for ¬ ⍱ X Y (boolean Not of Divisible) (new 2026-10-15)

U+2AEE

Combinatoric



Combinatoric —; dyadic ⋕ X Y: binomial X choose Y (new 2026-10-16)

U+2A69

Polynomials and Roots: Monadic

NextLocal



NextLocal — the Roots division step: given P, yields [R prime, P prime] where R prime is Local(P) and P prime is P deflated by synthetic division of R prime. (new 2026-10-19)

U+2113

Roots



Roots — sugar: ℞ P is ℜ ℓ P (Reveal NextLocal over P); yields all polynomial roots. (new 2026-10-18; factored via Reveal 2026-10-19)

U+211E

Differentiate



Differentiate — train verb ∂ T yields symbolically differentiated version of train T of coefficients, multiplying constants as needed; same order as Polynomial (new 2026-10-13)

U+2202

Integrate



Integrate — train verb ∫ T yields symbolically integrated version of train T of coefficients, dividing constants as needed; same order as Polynomial (new 2026-10-13)

U+222B

Solve

&

Solve — monadic & A: array → vector of polynomial coefficients; missing input row is all zeroes (new 2026-10-13)

U+0026

Optimize

𝔅

*Optimize —; monadic 𝔅 V: vector → polynomial; searches free variable with delta = 0.01 * domain, epsilon defaults to 1e-6*

U+214B

Local

𝔏

Local — monadic ⍵ V is 𝔅 ⫴𝔏 & V (FixedPoint Optimize Triple Solve V); approximates best x for a local minimal of V (new 2026-10-13)

U+1D50F

Polynomials and Roots: Dyadic

Reveal



Reveal — intrinsic. Dyadic ℜ P Y yields the successive application of P upon Y as Y shrinks to Nil. Epsilon from P has two elements: an alpha to apply P to next Y, and the next Y. (new 2026-10-18)

U+211C

Optimize



Optimize —; dyadic 𝔅 V ε: recurse into smaller domains until change in c < ε (new 2026-10-13)

U+214B

Integer Division



Integer Division — dyadic ⫽ X Y divides X by Y, yielding an integer (replaces Slash / integer division from manual p.19; / is now Reduce). Vector form: ⫽ V X where V is vector yields synthetic quotient of V by X (new 2026-10-14)

U+2AFD

Evaluate



Evaluate — dyadic 𝒇X T evaluates train T of coefficients upon X (new 2026-10-13)

U+1D58B

FixedPoint



Fixed point — dyadic ꞇø V guess: refine guess until epsilon < 1e-6 (max 100 steps); failure to converge returns infinity (new 2026-10-13)

U+AB44

Matrices: Niladic

Outer Product



Outer Product —; Niladic ⊗ yields 1 (new 2026-10-12)

U+2297

Inner Product



Inner Product —; Niladic · yields 1 (new 2026-10-12; matrix case 2026-10-14)

U+22C5

Multiply Matrices



Multiply Matrices —; Niladic ⋈ yields the 1×1 identity matrix (new 2026-10-14)

U+22C8

Matrices: Monadic

Outer Product



Outer Product —; Monadic ⊗ X yields X.

U+2297

Inner Product



Inner Product —; Monadic ⋅ X yields X.

U+22C5

Multiply Matrices



Multiply Matrices —; Monadic ⋈ X yields X.

U+22C8

L Factor



L Factor — monadic: the L (lower triangular, 1s on diagonal) factor of the LU decomposition of a rank-2 array (new 2026-10-14)

U+1D577

Determinant



Determinant — monadic ⌹X: determinant of square matrix X (new 2026-10-16)

U+233C

U Factor



U Factor — monadic: the U (upper triangular) factor of the LU decomposition of a rank-2 array (new 2026-10-14)

U+1D580

Matrices: Dyadic

Outer Product



Outer Product —; dyadic ⊗ X Y yields the outer product matrix where entry [i,j] is ⍺ X[i] Y[j]; shape is the concatenation of the shapes of X and Y.

U+2297

Inner Product



Inner Product —; dyadic ⋅ X Y: for vectors X, Y, the dot product (sum of pairwise products); for matrices, the Frobenius inner product: a scalar, the sum of all pairwise products.

U+22C5

Multiply Matrices



Multiply Matrices —; dyadic ⋈ X Y: matrix multiplication, (⋈ X Y)[i;j] = ⋅ X[i;] (⌽ Y)[j;]; inner dimensions must agree, non-square fine.

U+22C8

Arrays: Niladic

Pee

ℙ

Pee — Where⊡Pee yields Where [0, 1
(first two coordinates)

U+2119

Zee

ℤ

Zee — the last position of a slice
(Where⊡Zee yields Where [-1)

U+2124

Exx

𝕏

Exx — the zero position of a slice
(Where [0)

U+1D54F

Wye

𝕐

Wye — the midposition of a slice

U+1D550

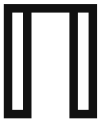
Extend



Extend —; niladic ⊢: sugar for ⊂ Nil

U+213E

Catenate



Catenate —; niladic ⊣: sugar for ⊂ Nil

U+213F

Grade Up



Grade Up — ⍋X yields the indices that sort X ascending. Niladic ⍋ yields 0 (from APL; new 2026-10-12)

U+234B

Grade Down



Grade Down — ⍒X yields the indices that sort X descending. Niladic ⍒ yields Nil (from APL; new 2026-10-12)

U+2352

Arrays: Monadic

Rank



Rank — monadic # X yields the number of dimensions in X; zero when scalar. (new 2026-10-18)

U+22D5

Permute



Permute —; monadic ⍨ X and dyadic X ⍨ Y work as in J's C.. Both modes intrinsic. (new 2026-10-19)

U+2A0A

Laminate



Laminate —; monadic ⍪ X: adds a leading singleton axis

U+2A5E

Anagram



Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)

U+1D49C

Enumerate

№

Enumerate — monadic № X: sugar for ⍕⍳≢⍴X; yields index-value pairs for X (new 2026-10-18)

U+2116

Reflect

÷

Reflect — monadic ÷X: sugar for ⍕⌽-⌽∘⍳X (new 2026-10-15)

U+223B

Count

77

⍳

U+2373

Kount

⍳̂

Kount — monadic ⍳̂B: takes a vector of T or Nil; returns a vector counting the T, turning each Nil to 0. Intrinsic. (new 2026-10-18)

U+2A23

⍳ Count

U+2373

Count — magical: when given a shape, fills an array matching shape. Scalar: behaves as normal. Array: fills array with next dimension limited by scalar from back rank — e.g. ⍳ 2 3 → rows of 0 1 2 (origin 0; last axis counts, outer axes replicate) (new 2026-10-11)

RavelRows



RavelRows — monadic ⍪X: yields the rows of X as a vector (Antitable; new 2026-10-15)

U+22F2

Ravel



Ravel — monadic ⍎X: flattens array X to vector, row-major (new 2026-10-14)

U+22F3

Reverse



Reverse —; monadic ⌽ X reverses elements along the last axis

U+233D

Transpose



Transpose — monadic ⍉ X reverses the order of axes (from APL; new 2026-10-12)

U+2349

Shape



Shape —; monadic ρ X: shape of X; scalar → 0; array → vector of dimensions; train → [1, length of train]; dictionary → [1, pairs present]

U+03C1

Extend



Extend —; monadic ⍐ X: Table X as in APL, adds trailing axis of length 1

U+213E

Catenate



Catenate —; monadic ⍪ X: catenates X to itself

U+213F

Rotate



Rotate —; monadic ⌽ X: rotate X toward origin by 1

U+29BD

Cardinality



Cardinality — "how big is this?"; scalar → -1; array → number of elements held; dictionary → number of pairs; train → length of train (new 2026-10-13)

U+2210

Center



Center — takes a shape, returns its integer halves expressed as a vector (e.g. [10, 6] → [5, 3]) (new 2026-10-13)

U+229E

Sight



Sight — takes 2D array, finds center, returns vector location of nearest non-zero scalar; ties pick one (new 2026-10-13)

U+2388

Arrays: Dyadic

Put



Put — ⫤⊨ X P A: value at coordinate P of array A becomes X (new 2026-10-18)

U+27DA

Permute



Permute —

U+2A0A

Laminate



Laminate —; dyadic ⩢ A B: yields [A; B]; when A and B are both scalars, yields A B (new 2026-10-17)

U+2A5E

Anagram



Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)

U+1D49C

Catenate



Catenate —; dyadic ⫿ X Y: catenate X and Y as in APL; variadic ⫿ A ... Z: catenate A through Z left to right (new 2026-10-14)

U+213F

Take



Take — ↑ X Y takes the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)

U+2191

Drop



Drop — ↓ X Y drops the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)

U+2193

Reverse



Reverse —; dyadic ⌽ X Y rotates Y by X positions (from APL; new 2026-10-12)

U+233D

Shape

ρ

Shape —; dyadic ρ S X: reshape X to shape S (per APL) (new 2026-10-13)

U+03C1

Extend



Extend —; dyadic ⍓ X N: extends by N axes (new 2026-10-14)

U+213E

Rotate



Rotate —; dyadic ⌽̂ X Y: rotate X by Y (Y positive toward origin, negative away) (new 2026-10-16)

U+29BD

Boxes: Niladic

Box



Box —; Niladic ⊂ yields Truth
(from APL Enclose; new 2026-10-12)

U+2282

Unbox



Unbox —; Niladic ⊃ yields Nil
(from APL Disclose; new 2026-10-12)

U+2283

Summarize



Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; niladic ⊚ is the 1,1 vector containing zero.

U+229A

Boxes: Monadic

Box



Box — monadic ⍈ A: wraps A in a box so it travels as a single element (new 2026-10-17)

U+2256

Unbox



Unbox — monadic ⍇ A: opens the box, yielding its contents (new 2026-10-17)

U+2257

Catalogue



Catalogue —; monadic ⍀ X: cartesian product of the boxed lists in vector X (new 2026-10-17)

U+2A37

Raze



Raze — monadic ⫻ A: sugar for Ravel Map Unbox A; opens every box and joins the contents into one vector (new 2026-10-17)

U+2AF5

Box



Box —; monadic ⊂X encloses X into a scalar box.

U+2282

Unbox



Unbox —; monadic ⊃X yields X.

U+2283

Summarize



Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar

U+229A

Zip

91



U+22B0



Zip

`U+22B0`

Zip — takes a train of vectors and zips them: ⍭ V1 V2 ... Vn yields an m×n matrix (m = vector length, n = train length); the resulting width is the length of the train; a scalar in the train expands into a column to fit the others; a container in the train contributes only its origin scalar, recursively (the summary effect); mismatched lengths Nil-pad (consistent with Diction); zips along the back rank for higher-rank arrays; ⍭ 1 2 yields the column [1;2] (new 2026-10-12; recycles the retired AntiSignum glyph)

Boxes: Dyadic

Summarize

94



U+229A

Fetch



Fetch — dyadic ⊸V X: applies V as search vector into X's items; follows path V down through nested boxes (new 2026-10-17)

U+22B8

Catalogue



Catalogue —; dyadic ⊗A B: yields [[A0,B0] ... [A1,B1]] ... [[AN,B0] ... [AN,BN]], the outer product matrix of pairs

U+2A37

Link



Link — dyadic ⫢X Y: sugar for Catenate Map Box X Y; boxes X and Y into a two-element list (new 2026-10-17)

U+2AE9



Summarize

U+229A

Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; Dyadic D ⊚ A: D (Direction) is a rank-1 array of coordinates; summarization starts at A[D], then proceeds as monadic; a scalar Direction s means start at (s, 0); Zip summarizes each train element (new 2026-10-12)

Box



Box —; Dyadic ⊂ X Y encloses X by Y levels; Y=0 leaves X unchanged.

U+2282

Unbox

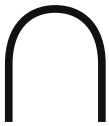


Unbox —; Dyadic ⊃ X Y picks element X from Y.

U+2283

Sets and Membership: Niladic

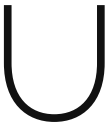
Intersection



Intersection —; niladic ∩: sugar for ⊂ Nil

U+2229

Union

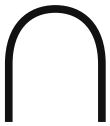


Union —; niladic ∪: sugar for ⊂ Nil

U+222A

Sets and Membership: Monadic

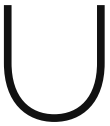
Intersection



Intersection —; monadic ∩ A: is A

U+2229

Union



Union —; monadic ∪X: unique as in APL

U+222A

InOf



InOf — the curry of In: ⇔InOf ⍷∈ X; monadic predicate testing membership in X. (new 2026-10-18)

U+22F8

Where



Where — monadic ⍸Y: returns vector of indices of all elements of array Y that are Truth (new 2026-10-18)

U+2A42

Enlist

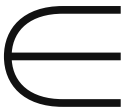


Enlist —; monadic ∈Y: as defined in APL (deep flatten to vector)

U+220A

Sets and Membership: Dyadic

In



In — dyadic ∈ Y X yields Truth when X is in Y, Nil otherwise (J's e.). (new 2026-10-18)

U+2208

Bucket



Bucket — dyadic ∐ X Y: yields index of interval in X that Y falls into (new 2026-10-18)

U+2210

InLast



InLast — dyadic ∈̇ X Y: yields the index of the last occurrence of X in Y; yields length of Y if not found (new 2026-10-18)

U+22F5

InAt



InAt — dyadic ⋲ X Y: yields the index of Y whose value equals X; yields length of Y if not found (new 2026-10-18)

U+22F9

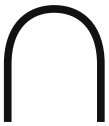
In



Enlist —; dyadic ∈ X Y: find/membership as defined in APL (new 2026-10-13)

U+220A

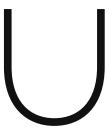
Intersection



Intersection —; dyadic ∩ A B: as in APL, elements in both (new 2026-10-14)

U+2229

Union



Union —; dyadic ∪ X Y: union as in APL (new 2026-10-14)

U+222A

Set Difference



Set Difference — dyadic ⍠A B: elements in A but not in B (new 2026-10-14)

U+2A43

Trains: Niladic

Reduce



Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Niladic / returns 0 (new 2026-10-12)

U+002F

Map



Map — ⍬ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Niladic ⍬ yields Truth, same as niladic 𝔗 (new 2026-10-12)

U+226C

Reduce (forward)



⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.

U+233F

Identity



Identity — 𝔗 V yields V unchanged; on a train, yields the train unchanged. Niladic 𝔗 yields Truth (new 2026-10-12)

U+1D574

Backstroke



Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Niladic \ yields 0 (new 2026-10-12)

U+005C

Stop



Stop — statement separator; separates statements on one line

U+22C4

Backstroke (forward)



Backstroke (forward) — Reverse Scan, paired with ⫻. ⫲ F V on a length-n V yields a length-n vector where element i is ⫻F (left fold) applied to the first i elements of V.

U+2340

Strip



Strip —; niladic: removes the train from head up to but not including the first Stop encountered

U+2ACF

Trains: Monadic

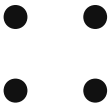
Meter

111



U+2214

Atomize



Atomize — monadic ∴A: reads up to the first 32 elements of A (origin to end) and prepares them for use by train elements ahead (new 2026-10-14)

U+2237

Duplicate



Duplicate — copies a train to a train twice as long: ⊷A B C → A B C A B C (was ==U+2A75, then ∴ U+2236 — settled 2026-10-12)

U+22B7

Caboose



Caboose — picks up the train from head to Stop, then converts boxed contents to a vector (inverse of Atomize) (new 2026-10-15)

U+2A78



Meter

U+2214

Meter — measures its argument: ∔ V is ≺ + ⊷ V (zip the needle with the signal); yields M, a sequence of (needle, signal) pairs, needle ∈ -/0/+. M works in place of V in all Collate modes: the needle selects the function, the function transforms the signal (was ±, corrected to ≺ 2026-10-12)

Reduce



Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Monadic / V yields a linear array of the scalars ripped from V, highest to lowest indices.

U+002F

Map



Map — ⍁ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Monadic ⍁ B is identical to Identity 𝔗 B.

U+226C

Reduce (forward)



⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.; Monadic ⌿ V and niladic ⌿ behave identically to / (new 2026-10-12)

U+233F

Zero



Zero — on a train, replaces all values with zeroes while keeping shapes (new 2026-10-12)

U+1D585

Backstroke

114

\

U+005C

Duplicate Head

Ⅱ

Duplicate Head — Ⅱ A ... yields A A ...; duplicates head of train, aka duplicate/dup (new 2026-10-14)

U+2161

Backstroke (forward)

⧵

Backstroke (forward) — Reverse Scan, paired with ⫻. ⧵ F V on a length-n V yields a length-n vector where element i is ⫻F (left fold) applied to the first i elements of V.; Monadic ⧵ V and niladic ⧵ behave identically to \ (new 2026-10-12)

U+2340

Pipeline



Pipeline —

U+2AF7



Backstroke

`U+005C`

Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Monadic \ V yields a linear array of the scalars ripped from V, lowest to highest indices.

Head



Head — sugar for ⊐1 (new 2026-10-15)

U+2991

Tail



Tail — sugar for ⊐-2 (new 2026-10-15)

U+2992

Duplicate Tail



Duplicate Tail — ⧟ ... A yields ... A A; duplicates tail of train (new 2026-10-13)

U+29DE

Strip

116



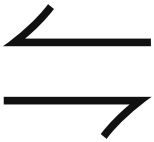
U+2ACF

⊐ Strip

U+2ACF

Strip —; monadic ⊐ X: removes the train from position X (0 is head) up to but not including the first Stop encountered; when X < 0, strips from position X in reverse order back to head, where −1 is the final tail position adjacent to the first Stop encountered (new 2026-10-15)

Swap



Swap — ⇋A B ... yields B A ...; swaps head two of train. Single: ⇋ A Stop X yields X. Bare: ⇋ Stop X yields X (new 2026-10-13)

U+21CB

Replicate

118



U+232C

Rotate Head



Rotate Head — ⟲A B C ... yields B C ... A; rotates head to tail (new 2026-10-13)

U+27F2

Rotate Tail



Rotate Tail — ⟳ ... A B C yields C ... A B; rotates tail to head (new 2026-10-13)

U+27F3



Replicate

U+232C

Replicate — ⌬ N X Stop yields X replicated N times; e.g. ⌬ 5 A Stop gives A A A A A. Zero case: ⌬ 0 A Stop X yields X. Negative N is antireplicate: ⌬ -1 A A A A A Stop yields A A A A. Empty antireplicate: ⌬ -1 Stop X yields X. Bare: ⌬ Stop X yields X (new 2026-10-13)

Nip



Nip — ↢ A B ... yields A ...; drops second, keeps head (new 2026-10-13)

U+21F7

Tuck



Tuck — ↣ A B ... yields A B A ...; copies head under second (new 2026-10-13)

U+21F8

Reverse Train



Reverse Train — reverses order of train (new 2026-10-13)

U+21F9

Over



Over — ⤺ A B ... yields B A B ...; copies second to head (new 2026-10-13)

U+293A

Triple

Ⅲ

Triple — Ⅲ X yields X X X; triplicates top of stack (new 2026-10-13)

U+2162

Fork



Fork — ⋔ F G H X: sugar for G H Swap F Dup X (J fork; new 2026-10-14)

U+22D4

Drop Train



Drop Train — sugar for reverse, replicate -1, reverse; drops head of train (new 2026-10-13)

U+291A

Trains: Dyadic

Applyself



Applyself — ⍣⍤1 f n a applies f n times to a; sugar for ⫷a ⍵⊂⍣⍺⍵ f n (from APL power; new 2026-10-12)

U+2363

Pipeline



Pipeline —; dyadic ⫷X Y applies vector Y, readable as a train of monadic functions, in sequence to X (new 2026-10-12)

U+2AF7

Functions: Monadic

Diction

¡

Diction — dict-backed decision-table constructor (was Vectorize, renamed 2026-10-11). ¡ A B → {A:B}; ¡ A B C D → {A:B,C:D}; monadic ¡ (a b c) self-maps into a set. Membership sugar: X member Y → ⊤(X ¿ (¡ Y))

U+00A1

Glyph

®

Glyph — the name for a single character value. Glyph ← ≡ ⌘: Glyph's function is the literalize train (via monadic ≡) — to make a Glyph, literalize

U+00AE

Decide

126

¿

U+00BF

Function

∇

Function — gets glyph del (∇, nabla): the function-definition operator. Monadic ∇Y executes array Y as an FSA. (monadic 2026-10-19)

U+2207

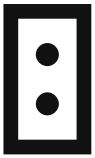


Decide

U+00BF

Decide — first-match generalized switch/match over n×2k train tables (was When, 2026-10-11). n×2 returns 1×1 (scalar); n×4 returns 1×2; generally n×2k returns 1×k. An explicit trailing C is unmatched-result filler, not an else branch; without filler, unmatched returns Nil without logging an error. Canonical: R ← X ¿ ¡ A B C

Print



Print — the quad-colon verb: prints the value from its right, then passes that value to its left (new 2026-10-11)

U+2360

(literalize)



Literalize next nonspace char, or ※0x???? by codepoint

U+2A33

Rail



Rail — monadic ⌐¯X yields a vector of symbols from train X (train-to-array conversion). (new 2026-10-19)

U+2ACD

Lay



Lay — monadic ¯¬V yields the train laid down from origin to limit of vector V (array-to-train, inverse of Rail). (new 2026-10-19)

U+2ACE

Lambda

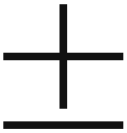


Lambda — lambda abstraction (new 2026-10-13)

U+039B

Functions: Dyadic

Collate



Collate — executes one of three trains upon a value: for less than zero, for zero, or for greater than zero (was AntiSignum ≺ U+22B0, renamed 2026-10-12; complement to signum)

U+00B1

Curry

131



U+1D49E



Curry

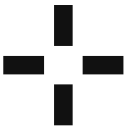
U+1D49E

Curry — intrinsic. Partial application: turns a dyadic verb into a monadic predicate by binding an argument, e.g. making = X from =. Semantics: Gets Q Curry P args: args to omega prime, omega prime shifts into alpha prime (binding the leftmost argument, textbook curry order), Q X makes X alpha, gets slice -1 of omega prime A, then call P; Q alpha yields P evaluated upon alpha prime, omega prime, and alpha as arguments in that order. Omega becomes nil upon Q's execution. Principle: Gets Q Curry P train is the principle for creating curries (Q is an example name assigned to the curried function, not a verb). (new 2026-10-18)

Errors: Niladic

Silence

135



U+22B9

Compress

136



U+22D6

Errline



Last error: which line

U+274C

Errmsg



Last error: what (type)

U+274E



Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Niladic ⊹ yields boxed Nil, a trial vector for enclosing operators to work on for starts (the dual of Compress; new 2026-10-12)



Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Niladic ⋖ yields boxed Truth, a trial vector for enclosing operators to work on for starts (a comprehension filter; new 2026-10-12)

Cut



Cut —; niladic: on error, erases the error and returns Nil

U+22EE

Debugger



✔ ⋄ *enters debugger iff error registers set*

U+2714

Errdmd



Last error: diamond position in line

U+2727

Succeed



Succeed — like Cut, except returns Truth (new 2026-10-15)

U+2999

Anticut



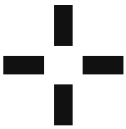
Anticut — on error throws to debugger; on return from debugger, returns Nil

U+22EF

Errors: Monadic

Silence

141



U+22B9

Compress

142



U+22D6

Cut



Cut —; monadic ⁝ X: checks X for Truth before returning

U+22EE



Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Monadic ⊹ V or ⊹ M means Z is identity: keep zeros (or zero-needle signals) as-is, drop nonzeros.



Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Monadic ⋖ V or ⋖ M means P is identity: pure filter, no transform.

Classes and Properties: Niladic

Constant

146

α

U+03B1

Epsilon

ε

Epsilon — holds the result of any getter, setter, or nabla when no explicit return glyph is used

U+03B5

Omega

ω

Omega — the vector of arguments passed to a setter (◅)

U+03C9

α Constant

U+03B1

Constant — declares a class constant: X α name ← value names a constant for class X. α is also the original (pre-override) value: read-only, may not be changed directly; trying registers an error. As the original value α is an ad hoc constant — contextual to the setter, not a declared class constant

Classes and Properties: Monadic

Increment



Increment —; monadic ⊕ D: invokes D's setter with nabla + 1 as the xthrement.

U+2295

Setter



Setter — the setter nabla: defines a property setter, as ∇ defines a function. Generalized: takes an xthrement argument — modifies instance data (increment/decrement generalized) rather than assigning (new 2026-10-14)

U+22B2

Getter



Getter — the getter nabla: defines a property getter, as ∇ defines a function

U+22B3

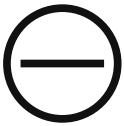
Property



Property —; Monadic ⋘ D: yields the result of D's getter (getter mode of D's class call) (new 2026-10-14)

U+22D8

Decrement



Decrement —; monadic ⊖ D: invokes D's setter with nabla - 1 as the xthrement.

U+2296

Divide By



Divide By —; monadic ⊘ D: invokes D's setter with nabla ÷ 2 as the xthrement (halve).

U+2298

Multiply By



Multiply By —; monadic ⊛ D: invokes D's setter with nabla × 2 as the xthrement (double).

U+229B

Classes and Properties: Dyadic

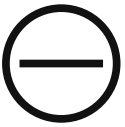
Increment



Increment —; Dyadic ⊕ D X: invokes D's setter with nabla + X as the xthrement (new 2026-10-14)

U+2295

Decrement



Decrement —; Dyadic ⊖ D X: invokes D's setter with nabla - X as the xthrement (new 2026-10-14)

U+2296

Multiply By



Multiply By —; Dyadic ⊛ D X: invokes D's setter with nabla ⌿ X as the xthrement (new 2026-10-14)

U+229B

Property



Property —; dyadic ⋘ I D: invokes D's getter (getter mode of D's class call), then invokes I's setter (setter mode of I's class call) with the result as argument; yields the assigned data as a pass-through.

U+22D8

Divide By



Divide By —; Dyadic ⊘ D X: invokes D's setter with nabla ÷ X as the xthrement (new 2026-10-14)

U+2298

(method binding)



Binds method to object: X⊡function

U+22A1

Gets



Gets — ⇔A G: A stored in symbols[G], G stored in instances[A]; bidirectional binding (new 2026-10-13)

U+2B04

The Virtual: Niladic

IntegerBounded



IntegerBounded — bounded integer class: min/max enforced on creation, assignment, math; clamp + silent sticky flag on violation (new 2026-10-15)

U+012A

Trapdoor



Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).

U+1D21D

VirtualSpace



VirtualSpace — the VR coordinate space: navel origin, forward half-sphere, [r;θ;φ], angles ±32767, distance in mm (new 2026-10-15)

U+1D4E5

Prim



Prim — base VR primitive: name, position, collision machinery; posted via Virtual (new 2026-10-16)

U+1D7D8

Raster



Raster — VR prim ⊑ Prim: array of pels, rank 4 [Y,X,1,3] (new 2026-10-16)

U+1D7D9

Point



Point — VR prim ⊑ Prim: dot prim, [N,6] (H,V,D,RGB) (new 2026-10-16)

U+1D7DA

Triangle



Triangle — VR prim ⊑ Prim: 9 points × 2 thetas + D + RGB = 22 numbers, [N,22] (new 2026-10-16)

U+1D7DB

The Virtual: Monadic

Trapdoor

159



U+1D21D

⊂ Trapdoor

`U+1D21D`

Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).; Monadic ⊂ A: puts the address of A into the accumulator before calling. ⇔ X ⊂ ∇ assembly ∇: X gets the address of the compiled assembly script. If X is so bound, niladic X calls the code. ε maps to the accumulator. The call yields ε

Dungeon

Dungeon: Places

Where

@

Where the hero is — bound to '@', the roguelike player glyph. Where⊡Zee yields Where [-1 (last); Where⊡Wye yields Where [1 (midposition — middle of 3 coords); Where⊡Exx yields Where [0 (zero). INDICES START AT ZERO

U+0040

Dungeon

£

The Dungeon — literally the set of all levels

U+00A3

Screen

§

The Screen — intrinsic 20×60 array (16 play + 4 I/O rows)

U+00A7

Level

¶

Level — the pilcrow: a single dungeon level, member of £ Dungeon

U+00B6

Cell

¢

A dungeon cell — the cent sign joins the family: § £ ¢. Cell⊡location ← £ Where: the cell's location is the Dungeon at Where. A dungeon square IS a dictionary of instances (the ⓘ registry's entries present at that square)

U+00A2

Up



Up — the upward/north movement. Where⊡up yields ∅ when Where⊡Zee is 0, else Where with Zee decremented (refined 2026-10-10: boundary-checked — top edge is a quiet Nil, not a negative index)

U+25B2

Down



Down — the downward/south movement, counterpart to Up. Where⊡down yields ∅ when Where⊡Zee ≥ Dungeon⊡Zee (bottom edge), else Where⊡Zee + 1 (confirmed 2026-10-10)

U+25BC

Dungeon: Classes

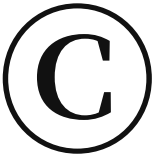
Class



Class — the circled dot. X⊡Class gets a RANK-2 ARRAY OF TRAINS: a class IS a decision table made of trains. Class instances are callable; the train defaults to Nil (∅)

U+2299

Classes



Classes — the set of all classes (plural of Class), parallel to £ Dungeon as the set of all levels. Implementation: a single twin (bidirectional word↔glyph) dictionary at global scope

U+24B8

Instances

167



U+24BE

Instance



Instance — the INDIVIDUAL instance (singular), distinct from Ⓘ the table. Joins the blackboard family (𝕏 𝕐 ℤ ℙ 𝕆). Carries the (glyph, hash-at-creation) identity tuple

U+1D540



Instances

U+24BE

Instances — the TABLE/set of all instances (plural), parallel to Ⓒ Classes. Implementation: a single twin (bidirectional word ↔ glyph) dictionary at global scope. Instance identity is the (glyph, hash-at-creation) tuple, used as the key field in both the global instance dictionary and each square (Cell) dictionary

Interpreter



Interpreter — the Wuxia interpreter (new 2026-10-15)

U+203D

Inherits



Inherits — the inheritance operator (square subset of)

U+2291

Isa



Isa — the is-a / instantiation operator (2026-10-10). Monadic ∈ Y: Y is a class to copy data from; new instance.ε = Y.ε. Dyadic X ∈ Y X: X is a class, optional initializer; new instance.ε = X.ε (new 2026-10-14)

U+22FF

Object



Object — a DUNGEON object, not a generic OOP object. The blackboard O joins the projection family (𝕏 𝕐 ℤ ℙ): the root of the dungeon's object hierarchy, what every dungeon entity inherits from

U+1D546

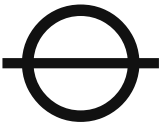
VirtualDistance



VirtualDistance — distance in VirtualSpace, 0-512 (new 2026-10-15)

U+27F0

Angle



Angle — a VR angle (theta/phi) (new 2026-10-15)

U+29B5

VirtualHorizontal



VirtualHorizontal — horizontal angle in VirtualSpace (new 2026-10-15)

U+29D3

VirtualVertical



VirtualVertical — vertical angle in VirtualSpace (new 2026-10-15)

U+29D7

Dungeon: Objects

Ring



Ring — the blackboard R: a dungeon object (ring), third of the dungeon entity glyphs

U+211D

Wand



Wand — left tack: a dungeon object (wand)

U+22A3

Armor



Armor — the blackboard A: a dungeon object (armor), first of the dungeon entity glyphs

U+1D538

Weapon



Weapon — the blackboard W: a dungeon object (weapon), second of the dungeon entity glyphs

U+1D54E

Amulet



Amulet — eight-petalled floret: a dungeon object (amulet). (Was U+1DF76, tofu on device — swapped 2026-10-10)

U+2741

Food



Food — the blackboard F: a dungeon object (food)

U+1D53D

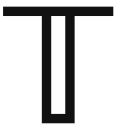
Scroll



Scroll — the blackboard S: a dungeon object (scroll)

U+1D54A

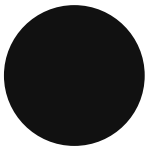
Tool



Tool — the blackboard T: a dungeon object (tool)

U+1D54B

Gem



Gem — solid black circle: a dungeon object (gem). (Was U+2733 — swapped 2026-10-10)

U+25CF

Potion



Potion — heavy exclamation: a dungeon object (potion)

U+2757

Dungeon: Entities

Player



Player — black smiling face: the dungeon's player entity. Inherits Monster plus a few extra fields Monster does not have

U+263B

Monster



Monster — the blackboard M: a dungeon monster, first of the monster glyphs

U+1D544

Dungeon: Food

Tin



Tin — circled white star: a dungeon food item (tin), first of the object species

U+272A

RefriedBeans



RefriedBeans — dark-encircled six: grants teleportation

U+277B

Spinach



Spinach — dark-encircled seven: Popeye strength

U+277C

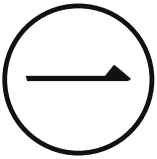
Sushi



Sushi — circled congratulation: the sushi/sashimi food category. (Was U+2722 — swapped 2026-10-11)

U+3297

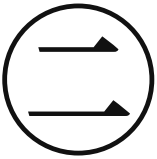
CarpSashimi



CarpSashimi — circled kanji: sushi item

U+3280

WhaleSashimi



WhaleSashimi — circled kanji: sushi item

U+3281

EelSashimi



EelSashimi — circled kanji: sushi item

U+3282

SalmonSashimi



SalmonSashimi — circled kanji: sushi item

U+3283

TunaSashimi



TunaSashimi — circled kanji: sushi item

U+3284

CrabSashimi



CrabSashimi — circled kanji: sushi item

U+3285

DragonSashimi



DragonSashimi — circled kanji: sushi item

U+3286

TurtleSashimi



TurtleSashimi — circled kanji: sushi item

U+3287

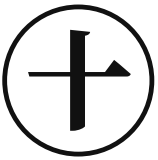
Oyster



Oyster — circled kanji: sushi item

U+3288

Uni



Uni — circled kanji: sushi item

U+3289

Gyuniku



Gyuniku — circled kanji: sushi item

U+328A

Asian



Asian — squared hand: the Asian food category. (Was U+328C — swapped 2026-10-11)

U+1F210

Ramen



Ramen — squared cjk unified ideograph-5b57: Asian food

U+1F211

FortuneCookie



FortuneCookie — squared cjk unified ideograph-53cc: Asian food

U+1F212

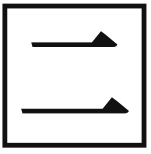
Kelp



Kelp — squared katakana de: Asian food

U+1F213

Dumplings



Dumplings — squared cjk unified ideograph-4e8c: Asian food

U+1F214

BowlOfRice



BowlOfRice — squared cjk unified ideograph-591a: Asian food

U+1F215

SpringRolls



SpringRolls — squared cjk unified ideograph-89e3: Asian food

U+1F216

Tofu



Tofu — squared cjk unified ideograph-5929: Asian food

U+1F217

PreservedEgg



PreservedEgg — squared cjk unified ideograph-4ea4: Asian food

U+1F218

Soup



Soup — circled twenty: the soup category

U+2473

Deli



Deli — black diamond minus white X: the delicatessen food category (the 25 staples, names lost 2026-10-10 — rebuild pending)

U+2756

Fruit



Fruit — domino tile back: the fruit food category

U+1F030

CantoneseDuck



CantoneseDuck — squared cjk unified ideograph-6620: Asian food

U+1F219

Egg

Ø

Egg — o with stroke: the egg. (Was U+2800 — swapped 2026-10-11)

U+00F8

Meat



Meat — CJK beautiful: the meat category

U+4E3D

Creature



Creature — o with long stroke overlay: the monster-shuffle category

U+A74B

Candy



Candy — mahjong joker: the candy category

U+1F02A

Goodbar



Goodbar — mahjong east wind: candy

`U+1F000`

Honee



Honee — mahjong south wind: candy

`U+1F001`

DRation



DRation — mahjong west wind: candy

`U+1F002`

DoveBar



DoveBar — mahjong north wind: candy

`U+1F003`

AcidBroth

❶

AcidBroth — dark-encircled number: soup

U+2776

GoldenSoup

❷

GoldenSoup — dark-encircled number: soup

U+2777

StonedSoup

❸

StonedSoup — dark-encircled number: soup

U+2778

CrowBar



CrowBar — mahjong winter: candy

U+1F029



MoldySoup

MoldySoup — dark-encircled number: soup

U+2779



Manwich

Manwich — dark-encircled number: soup

U+277A



CreamOfAlgaeSoup

CreamOfAlgaeSoup — dark-encircled number: soup

U+277E



ChickenNoodleSoup

ChickenNoodleSoup — dark-encircled number: soup

U+277F

Apple



Apple — domino tile: fruit, inherits Fruit

U+1F032

Orange



Orange — domino tile: fruit, inherits Fruit

U+1F033

Pear



Pear — domino tile: fruit, inherits Fruit

U+1F034

Melon



Melon — domino tile: fruit, inherits Fruit

U+1F035

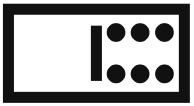
Banana



Banana — domino tile: fruit, inherits Fruit

U+1F036

Carrot



Carrot — domino tile: fruit, inherits Fruit

U+1F037

Fig



Fig — domino tile: fruit, inherits Fruit

U+1F038

Lemon



Lemon — domino tile: fruit, inherits Fruit

U+1F039

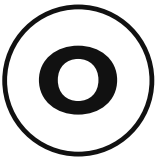
Tomb



Tomb — circled 10 on black square: safe-row tin, now Sundry

U+3248

Sundry



Sundry — circled ideograph: the sundry/miscellaneous category

U+3267

Lime



Lime — domino tile: fruit, inherits Fruit

U+1F03A

Grapes



Grapes — domino tile: fruit, inherits Fruit

U+1F03B

Pesto



Pesto — circled 20 on black square: safe-row tin, now Sundry

U+3249

KelpFrond



KelpFrond — circled water: sundry item (kelp is seaweed)

U+328C

EucalyptusLeaf



EucalyptusLeaf — circled wood: sundry item

U+328D

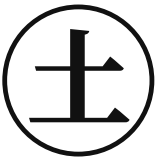
CandyBar



CandyBar — circled metal: sundry item

U+328E

LembasWafer



LembasWafer — circled earth: sundry item

U+328F

CramRation



CramRation — circled sun: sundry item

U+3290

FoodRation



FoodRation — circled medicine: sundry item

U+32A9

KRation



KRation — circled religion: sundry item

U+32AA

CloveOfGarlic



CloveOfGarlic — circled attention: sundry item

U+329F

SprigOfWolfsbane



SprigOfWolfsbane — circled item: sundry item

U+32A0

LumpOfRoyalJelly



LumpOfRoyalJelly — circled copy: sundry item

U+32A2

CRation



CRation — circled study: sundry item

U+32AB

Dungeon: Species inheritance

EarthElemental



⊑ Elementals — Species inherits its category (2026-10-10)

U+2295

FireElemental



⊑ Elementals — Species inherits its category (2026-10-10)

U+229B

AirElemental



⊑ Elementals — Species inherits its category (2026-10-10)

U+2A37

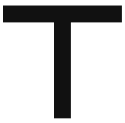
WaterElemental



⊑ Elementals — Species inherits its category (2026-10-10)

U+2A38

PurpleWorm



⊑ Worms — Species inherits its category (2026-10-10)

U+22A4

BabyPurpleWorm



⊑ Worms — Species inherits its category (2026-10-10)

U+22A6

LongWorm



⊑ Worms — Species inherits its category (2026-10-10)

U+2A53

BabyLongWorm



⊑ Worms — Species inherits its category (2026-10-10)

U+2A55

LargeKobold



⊑ Kobolds — Species inherits its category (2026-10-10)

U+049C

Kobold



⊑ Kobolds — Species inherits its category (2026-10-10)

U+049D

MasterMindFlayer



⊑ Mindflayers — Species inherits its category (2026-10-10)

U+04F4

MindFlayer



⊑ Mindflayers — Species inherits its category (2026-10-10)

U+04F5

StoneGiant

Ġ

⊑ Giants — Species inherits its category (2026-10-10)

U+0120

Giant

Ģ

⊑ Giants — Species inherits its category (2026-10-10)

U+0122

KoboldLord

Ќ

⊑ Kobolds — Species inherits its category (2026-10-10)

U+040C

KoboldShaman

Ҝ

⊑ Kobolds — Species inherits its category (2026-10-10)

U+049B

FireGiant

Ĝ

⊑ Giants — Species inherits its category (2026-10-10)

`U+011C`

HillGiant

Ğ

⊑ Giants — Species inherits its category (2026-10-10)

`U+011E`

FrostGiant

Ǵ

⊑ Giants — Species inherits its category (2026-10-10)

`U+0193`

Ettin

Gˇ

⊑ Giants — Species inherits its category (2026-10-10)

`U+01E6`

VampireBat

ƀ

⊑ Vampires — Species inherits its category (2026-10-10)

U+0180

StormGiant

Ǧ

⊑ Giants — Species inherits its category (2026-10-10)

U+01E4

Vampire

Ỿ

⊑ Vampires — Species inherits its category (2026-10-10)

U+1E7F

Titan

⅁

⊑ Giants — Species inherits its category (2026-10-10)

U+2141

VampireLeader



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+1E7D

VampireLady



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

VampireLord



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

Nosferatu



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

HumanMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

ElfMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

DwarfMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

GnomeMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

OrcMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

KoboldMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

GiantMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

EttinMummy



⊑ Undead — Species inherits its category (2026-10-10)

U+046A

KoboldZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

GnomeZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

OrcZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

DwarfZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

ElfZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

HumanZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

GiantZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

EttinZombie



⊏ *Undead — Species inherits its category (2026-10-10)*

U+2124

Ghoul

g

⊑ Undead — Species inherits its category (2026-10-10)

U+210A

WoodElf



⊑ Elves — Species inherits its category (2026-10-10)

U+2618

GreenElf



⊑ Elves — Species inherits its category (2026-10-10)

U+2618

GreyElf



⊑ Elves — Species inherits its category (2026-10-10)

U+2618

BabyDragon

⁇

⊏ Dragons — Species inherits its category (2026-10-10)

U+2047

Elf



⊏ Elves — Species inherits its category (2026-10-10)

U+2618

ElfNoble



⊏ Elves — Species inherits its category (2026-10-10)

U+2E19

ElvenMonarch

支

⊏ Elves — Species inherits its category (2026-10-10)

U+2F40

Centipede

¢

⊑ Spiders — Species inherits its category (2026-10-10)

U+00A2

CaveSpider

¤

⊑ Spiders — Species inherits its category (2026-10-10)

U+00A4

Scorpion

±

⊑ Spiders — Species inherits its category (2026-10-10)

U+00B1

Dragon

⁇

⊑ Dragons — Species inherits its category (2026-10-10)

U+2047

GiantSpider

Ø

⊑ Spiders — Species inherits its category (2026-10-10)

U+00D8

Mastodon

Ψ

⊑ Beasts — Species inherits its category (2026-10-10)

U+03A8

Mumak

Ψ

⊑ Beasts — Species inherits its category (2026-10-10)

U+03A8

Baluchitherium

ϒ

⊑ Beasts — Species inherits its category (2026-10-10)

U+03D2

Wumpus

¿

⊑ Beasts — Species inherits its category (2026-10-10)

U+00BF

Trapper



⊑ Beasts — Species inherits its category (2026-10-10)

U+048A

Minotaur

Ṁ

⊑ Beasts — Species inherits its category (2026-10-10)

U+1E40

Titanothere



⊑ Beasts — Species inherits its category (2026-10-10)

U+213A

Bugbear

ö

⊑ Beasts — Species inherits its category (2026-10-10)

U+00F6

Owlbear

ø

⊑ Beasts — Species inherits its category (2026-10-10)

U+00F8

Jabberwock

ƹ

⊑ Beasts — Species inherits its category (2026-10-10)

U+01B8

Rothe

ȣ

⊑ Beasts — Species inherits its category (2026-10-10)

U+0222

Shade

@

⊑ Guises — Species inherits its category (2026-10-10)

U+0040

Leocrotta

ꝉ

⊑ Beasts — Species inherits its category (2026-10-10)

U+023D

Stalker

Ǝ

⊑ Lurkers — Species inherits its category (2026-10-10)

U+042D

LurkerAbove



⊑ Lurkers — Species inherits its category (2026-10-10)

U+214F

Skeleton



⊑ Guises — Species inherits its category (2026-10-10)

U+0040

Foocubus



⊑ Guises — Species inherits its category (2026-10-10)

U+0040

MountainNymph



⊑ Guises — Species inherits its category (2026-10-10)

U+0040

WoodNymph



⊑ Guises — Species inherits its category (2026-10-10)

U+0040

WaterNymph

@

⊑ Guises — Species inherits its category (2026-10-10)

U+0040

MerfolkRoyal

⁇

⊑ Merfolk — Species inherits its category (2026-10-10)

U+2047

MerfolkNoble

⁇

⊑ Merfolk — Species inherits its category (2026-10-10)

U+2047

MerfolkWitch

⁇

⊑ Merfolk — Species inherits its category (2026-10-10)

U+2047

WizardOfYendor

¥

⊑ Specials — Species inherits its category (2026-10-10)

U+00A5

Shopkeeper

§

⊑ Specials — Species inherits its category (2026-10-10)

U+00A7

MerfolkCommon

⁇

⊑ Merfolk — Species inherits its category (2026-10-10)

U+2047

GrandMaster

⁇

⊑ Specials — Species inherits its category (2026-10-10)

U+2047

Tin



⊏ Food — Tin inherits Food — first object species joins the hierarchy (2026-10-10)

U+272A

AcidBroth



⊏ Soup — Soup inherits Soup (2026-10-10)

U+2776

GoldenSoup



⊏ Soup — Soup inherits Soup (2026-10-10)

U+2777

StonedSoup



⊏ Soup — Soup inherits Soup (2026-10-10)

U+2778

4

MoldySoup

⊑ Soup — Soup inherits Soup (2026-10-10)

U+2779

5

Manwich

⊑ Soup — Soup inherits Soup — the human-flesh tin (2026-10-10)

U+277A

9

CreamOfAlgaeSoup

⊑ Soup — Soup inherits Soup (2026-10-10)

U+277E

10

ChickenNoodleSoup

⊑ Soup — Soup inherits Soup (2026-10-10)

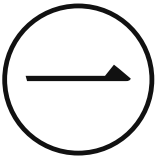
U+277F



RefriedBeans

⊑ Soup — Soup item inherits Soup — grants teleportation (2026-10-11)

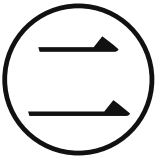
U+277B



CarpSashimi

⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)

U+3280



WhaleSashimi

⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)

U+3281



Sushi

⊑ Food — Sushi inherits Food — the sushi food category (2026-10-10)

U+3297

EelSashimi



㆓ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3282

SalmonSashimi



㆓ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3283

TunaSashimi



㆓ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3284

CrabSashimi



㆓ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3285

DragonSashimi



⿲ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3286

TurtleSashimi



⿲ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3287

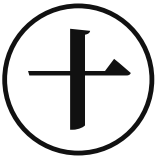
Oyster



⿲ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3288

Uni



⿲ Sushi — Individual sushi
inherits Sushi (2026-10-10)

U+3289

Gyuniku



⊑ Sushi — Individual sushi inherits Sushi (2026-10-10)

U+328A

Asian



⊑ Food — Asian inherits Food — the Asian food category (2026-10-10)

U+1F210

Ramen



⊑ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F211

FortuneCookie



⊑ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F212

Kelp



⿺ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F213

Dumplings



⿺ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F214

BowlOfRice



⿺ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F215

SpringRolls



⿺ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F216

Deli



⿸ Food — Deli inherits Food — the delicatessen food category (2026-10-10)

U+2756

Tofu



⿸ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F217

PreservedEgg



⿸ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F218

CantoneseDuck



⿸ Asian — Individual Asian food inherits Asian (2026-10-10)

U+1F219

KelpFrond



⿸ Sundry — Sundry item inherits Sundry (2026-10-10)

U+328C

EucalyptusLeaf



⿸ Sundry — Sundry item inherits Sundry (2026-10-10)

U+328D

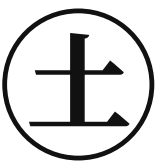
CandyBar



⿸ Sundry — Sundry item inherits Sundry (2026-10-10)

U+328E

LembasWafer



⿸ Sundry — Sundry item inherits Sundry (2026-10-10)

U+328F

㊐

CramRation

㊐ Sundry — Sundry item inherits Sundry (2026-10-10)

U+3290

㊐

FoodRation

㊐ Sundry — Sundry item inherits Sundry (2026-10-10)

U+32A9

㊐

KRation

㊐ Sundry — Sundry item inherits Sundry (2026-10-10)

U+32AA

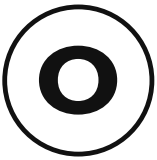
㊐

CRation

㊐ Sundry — Sundry item inherits Sundry (2026-10-10)

U+32AB

Sundry



㆓ Deli — Sundry is part of Deli (2026-10-10)

U+3267

CloveOfGarlic



㆓ Sundry — Sundry item inherits Sundry (2026-10-10)

U+329F

SprigOfWolfsbane



㆓ Sundry — Sundry item inherits Sundry (2026-10-10)

U+32A0

LumpOfRoyalJelly



㆓ Sundry — Sundry item inherits Sundry (2026-10-10)

U+32A2

Fruit



⊑ Food — Fruit inherits Food — the fruit food category (2026-10-10)

U+1F030

Apple



⊑ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F032

Orange



⊑ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F033

Pear



⊑ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F034

Melon



⊑ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F035

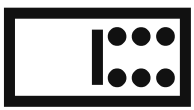
Banana



⊑ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F036

Carrot



⊑ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F037

Fig



⊑ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F038

Soup



⊏ Tin — Soup inherits Tin — the soup category (2026-10-11)

U+2473

Lemon



⊏ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F039

Lime



⊏ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F03A

Grapes



⊏ Fruit — Fruit item inherits Fruit (2026-10-10)

U+1F03B

Goodbar



⊑ Candy — Goodbar inherits Candy (2026-10-11)

U+1F000

Honee



⊑ Candy — Honee inherits Candy (2026-10-11)

U+1F001

DRation



⊑ Candy — DRation inherits Candy (2026-10-11)

U+1F002

Candy



⊑ Food — Candy inherits Food — the candy category (2026-10-11)

U+1F02A

Tomb



⿺ Sundry — Tomb inherits Sundry
(2026-10-11)

U+3248

Pesto



⿺ Sundry — Pesto inherits Sundry
(2026-10-11)

U+3249

DoveBar



⿺ Candy — DoveBar inherits Candy
(2026-10-11)

U+1F003

CrowBar



⿺ Candy — CrowBar inherits Candy
(2026-10-11)

U+1F029

Egg

Ø

⊑ Creature — Egg inherits Creature (2026-10-11)

U+00F8

Spinach



⊑ Soup — Spinach (NetHack tin filling, Popeye strength) inherits Soup (2026-10-11)

U+277C

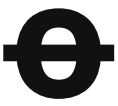
Meat



⊑ Creature — Meat inherits Creature — the meat category (2026-10-11)

U+4E3D

Creature



⊑ Tin — Creature inherits Tin — the creature-meat tin category; the category that shuffles monsters as consumables (tins of monster meat) (2026-10-11)

U+A74B

IntegerBounded

Ī

⊑ *Integer — IntegerBounded inherits Integer (new 2026-10-15)*

U+012A

IntegerBounded.max

⁇

⊑ *Integer — IntegerBounded.max is an Integer (new 2026-10-15)*

U+2047

IntegerBounded.min

⁇

⊑ *Integer — IntegerBounded.min is an Integer (new 2026-10-15)*

U+2047

IntegerBounded.epsilon

⁇

⊑ *Integer — IntegerBounded.epsilon is an Integer (new 2026-10-15)*

U+2047

IntegerBounded.nabla

⁇

IntegerBounded.nabla enforces bounded gets and sets (new 2026-10-15)

U+2047

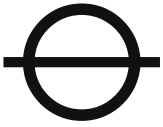
IntegerBounded.nabla

⁇

⊑ Interpreter — IntegerBounded.nabla is an Interpreter (new 2026-10-15)

U+2047

Angle



⊑ IntegerBounded — Angle inherits IntegerBounded (new 2026-10-15)

U+29B5

VirtualHorizontal



⊑ Angle — VirtualHorizontal inherits Angle (new 2026-10-15)

U+29D3

Egg

Ø

Egg stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)

U+00F8

VirtualDistance



⊑ IntegerBounded — VirtualDistance inherits IntegerBounded (new 2026-10-15)

U+27F0

VirtualVertical



⊑ Angle — VirtualVertical inherits Angle (new 2026-10-15)

U+29D7

Meat



Meat stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)

U+4E3D

Index

English Index

AcidBroth 189-1
AcidBroth 220-2
AirElemental 199-3
Amulet 173-1
Anagram 75-4
Anagram 82-4
Angle 169-2
Angle 238-3
Anticut 138-1
Antilog 28-2
Apple 191-1
Apple 231-2
Applyself 122-1
Approx 13-3
Arccosine 26-1
Arcsine 26-2
Arctangent 26-4
Armor 172-3
Asian 183-4
Asian 225-2
Atomize 110-2
BabyDragon 211-1
BabyLongWorm 200-4
BabyPurpleWorm 200-2
Backstroke 108-1
Backstroke 113-1
Backstroke (forward) 108-3
Backstroke (forward) 113-3
Baluchitherium 213-4
Banana 192-1
Banana 232-2
Base Ten Log 29-4
Bell 39-3
Bell 44-3
Bell 47-1
BowlOfRice 185-1
BowlOfRice 226-3
Box 87-1
Box 89-1
Box 90-1
Box 95-1
Bucket 103-2
Bugbear 215-1
Caboose 110-4
Candy 187-4
Candy 234-4
CandyBar 194-4
CandyBar 228-3
CantoneseDuck 186-4
CantoneseDuck 227-4
Cardinality 80-1
CarpSashimi 181-1
CarpSashimi 222-2
Carrot 192-2
Carrot 232-3
Catalogue 89-3
Catalogue 93-3
Catenate 73-2
Catenate 79-3
Catenate 83-1
CaveSpider 212-2
Ceiling 25-1
Ceiling 35-2
Cell 163-1
Center 80-2
Centipede 212-1
ChickenNoodleSoup 190-4
ChickenNoodleSoup 221-4
Class 166-1
Classes 166-2
CloveOfGarlic 196-1
CloveOfGarlic 230-2
Collate 130-1
Combinatoric 51-3
Combinatoric 54-1
Compress 134-2

Compress 140-2
Congruent 9-3
Congruent 11-3
Conjugate 21-3
Constant 145-1
Cosine 25-4
Count 76-3
CrabSashimi 182-2
CrabSashimi 223-4
CramRation 195-2
CramRation 229-1
CRation 196-4
CRation 229-4
CreamOfAlgaeSoup 190-3
CreamOfAlgaeSoup 221-3
Creature 187-3
Creature 236-4
CrowBar 189-4
CrowBar 235-4
Curry 130-2
Cut 137-1
Cut 140-3
D10 42-1
D2 41-2
D4 41-3
D6 41-4
Deal 45-1
Debugger 137-2
Decide 125-3
Decrement 149-1
Decrement 151-2
Deli 186-2
Deli 227-1
Determinant 67-1
Dice 40-4
Dice 44-4
Dice 47-2
Dice Percentile 41-1
Diction 125-1
Differentiate 57-3
Divide (Complex) 18-3
Divide (Complex) 23-3
Divide (Complex) 33-3
Divide (Polar) 18-4
Divide (Polar) 23-4
Divide (Polar) 33-4
Divide By 149-2
Divide By 152-1
Divisible 50-1
Divisible 53-2
Double 21-4
DoveBar 188-4
DoveBar 235-3
Down 163-3
Dragon 212-4
DragonSashimi 182-3
DragonSashimi 224-1
DRation 188-3
DRation 234-3
Drop 83-3
Drop Train 120-3
Dumplings 184-4
Dumplings 226-2
Dungeon 162-2
Duplicate 110-3
Duplicate Head 113-2
Duplicate Tail 115-3
DwarfMummy 206-3
DwarfZombie 208-4
EarthElemental 199-1
EelSashimi 181-3
EelSashimi 223-1
Egg 187-1
Egg 236-1
Egg 239-1
Elf 211-2
ElfMummy 206-2
ElfNoble 211-3
ElfZombie 209-1
ElvenMonarch 211-4
Enlist 101-1
Enumerate 76-1
Epsilon 145-2
Equality 12-1

Errdmd 137-3
Errline 134-3
Errmsg 134-4
Ettin 203-4
EttinMummy 207-4
EttinZombie 209-4
EucalyptusLeaf 194-3
EucalyptusLeaf 228-2
Euler 40-3
Evaluate 60-4
Extend 73-1
Extend 79-2
Extend 84-2
Exx 72-3
Factor 51-2
Factorial 19-1
Factorial 29-1
Factorial 35-1
Fetch 93-2
Fig 192-3
Fig 232-4
FireElemental 199-2
FireGiant 203-1
FixedPoint 61-1
Floor 25-2
Floor 35-3
Foocubus 217-2
Food 173-2
FoodRation 195-3
FoodRation 229-2
forall 13-2
Fork 120-2
Fornone 14-2
Forsome 14-1
FortuneCookie 184-2
FortuneCookie 225-4
Fourier 29-2
FrostGiant 203-3
Fruit 186-3
Fruit 231-1
Function 125-4
Gamma 40-1
GCD 53-3
Gem 174-1
Gets 152-3
Getter 148-3
Ghoul 210-1
Giant 202-2
GiantMummy 207-3
GiantSpider 213-1
GiantZombie 209-3
Glyph 125-2
GnomeMummy 206-4
GnomeZombie 208-2
Golden 40-2
GoldenSoup 189-2
GoldenSoup 220-3
Goodbar 188-1
Goodbar 234-1
Grade Down 73-4
Grade Up 73-3
GrandMaster 219-4
Grapes 193-4
Grapes 233-4
Greater Or Equal 12-4
Greater Than 11-2
GreenElf 210-3
GreyElf 210-4
Gyuniku 183-3
Gyuniku 225-1
Half 39-1
Half 44-1
Head 115-1
HillGiant 203-2
Honee 188-2
Honee 234-2
HumanMummy 206-1
HumanZombie 209-2
Hyperbolic Arccosine 28-4
Hyperbolic Arcsine 27-3
Hyperbolic Arctangent 28-3
Hyperbolic Cosine 27-2
Hyperbolic Sine 27-1
Hyperbolic Tangent 27-4

Identity 107-4
Implies 13-1
In 103-1
In 104-1
InAt 103-4
Increment 148-1
Increment 151-1
Indivisible 50-2
Indivisible 53-4
Inherits 168-2
InLast 103-3
Inner Product 64-2
Inner Product 66-2
Inner Product 69-2
InOf 100-3
Instance 166-4
Instances 166-3
Integer Division 60-3
IntegerBounded 155-1
IntegerBounded 237-1
IntegerBounded.epsilon 237-4
IntegerBounded.max 237-2
IntegerBounded.min 237-3
IntegerBounded.nabla 238-1
IntegerBounded.nabla 238-2
Integrate 57-4
Interpreter 168-1
Intersection 98-1
Intersection 100-1
Intersection 104-2
InverseFourier 29-3
Invert 9-1
Isa 168-3
IsAt 11-4
Jabberwock 215-3
Kelp 184-3
Kelp 226-1
KelpFrond 194-2
KelpFrond 228-1
Kobold 201-2
KoboldLord 202-3
KoboldMummy 207-2
KoboldShaman 202-4
KoboldZombie 208-1
Kount 76-4
KRation 195-4
KRation 229-3
L Factor 66-4
Lambda 128-1
Laminate 75-3
Laminate 82-3
LargeKobold 201-1
Lay 127-4
LCM 53-1
LembasWafer 195-1
LembasWafer 228-4
Lemon 192-4
Lemon 233-2
Leocrotta 216-2
Less Or Equal 12-3
Less Than 11-1
Level 162-4
Lime 193-3
Lime 233-3
Link 93-4
(literalize) 127-2
Local 58-3
LongWorm 200-3
LumpOfRoyalJelly 196-3
LumpOfRoyalJelly 230-4
LurkerAbove 216-4
Magnitude 30-1
Manwich 190-2
Manwich 221-2
Map 107-2
Map 112-2
MasterMindFlayer 201-3
Mastodon 213-2
Meat 187-2
Meat 236-3
Meat 239-4
Melon 191-4
Melon 232-1
MerfolkCommon 219-3

MerfolkNoble 218-3
MerfolkRoyal 218-2
MerfolkWitch 218-4
Meter 110-1
(method binding) 152-2
MindFlayer 201-4
Minotaur 214-3
Minus (Complex) 18-1
Minus (Complex) 23-1
Minus (Complex) 33-1
MoldySoup 190-1
MoldySoup 221-1
Monster 177-2
MountainNymph 217-3
Multiply 19-4
Multiply 24-4
Multiply 34-4
Multiply By 149-3
Multiply By 151-3
Multiply Matrices 64-3
Multiply Matrices 66-3
Multiply Matrices 69-3
Mumak 213-3
Natural Log 28-1
Natural Log 35-4
Negate 21-2
Negate / Subtract 17-2
NextLocal 57-1
Nil 7-2
Nip 119-1
Nosferatu 205-4
Not 9-2
Object 168-4
Omega 145-3
Optimize 58-2
Optimize 60-2
Orange 191-2
Orange 231-3
OrcMummy 207-1
OrcZombie 208-3
Outer Product 64-1
Outer Product 66-1
Outer Product 69-1
Over 119-4
Owlbear 215-2
Oyster 183-1
Oyster 224-3
Pear 191-3
Pear 231-4
Pee 72-1
Percent 19-2
Percent 24-1
Percent 34-1
Permute 75-2
Permute 82-2
Pesto 194-1
Pesto 235-2
Pi 39-2
Pick 42-2
Pick 44-2
Pipeline 113-4
Pipeline 122-2
Player 177-1
Plus 32-1
Plus (Complex) 17-3
Plus (Complex) 22-3
Plus (Complex) 32-3
Point 156-2
Potion 174-2
PreservedEgg 185-4
PreservedEgg 227-3
Prim 155-4
Prime 50-4
Prime 51-1
Primes 50-3
Print 127-1
Property 148-4
Property 151-4
PurpleWorm 200-1
Put 82-1
Quotient 24-3
Quotient 34-3
Rail 127-3
Ramen 184-1

Ramen 225-3
Rank 75-1
Raster 156-1
Ravel 78-2
RavelRows 78-1
Raze 89-4
Real 39-4
Reduce 107-1
Reduce 112-1
Reduce (forward) 107-3
Reduce (forward) 112-3
Reflect 76-2
RefriedBeans 180-2
RefriedBeans 222-1
Replicate 117-2
Reveal 60-1
Reverse 78-3
Reverse 83-4
Reverse Train 119-3
Ring 172-1
Roots 57-2
Rotate 79-4
Rotate 84-3
Rotate Head 117-3
Rotate Tail 117-4
Rothe 215-4
SalmonSashimi 181-4
SalmonSashimi 223-2
Scorpion 212-3
Screen 162-3
Scroll 173-3
Set Difference 104-4
Setter 148-2
Shade 216-1
Shape 79-1
Shape 84-1
Shopkeeper 219-2
Sight 80-3
Signum 21-1
Signum / Plus 17-1
Silence 134-1
Silence 140-1
Sine 25-3
Skeleton 217-1
Solve 58-1
Soup 186-1
Soup 233-1
Spinach 180-3
Spinach 236-2
SprigOfWolfsbane 196-2
SprigOfWolfsbane 230-3
SpringRolls 185-2
SpringRolls 226-4
Square 19-3
Square 36-1
Square Root 22-2
Stalker 216-3
Star 24-2
Star 34-2
StonedSoup 189-3
StonedSoup 220-4
StoneGiant 202-1
Stop 108-2
StormGiant 204-2
Strip 108-4
Strip 115-4
Subtract 32-2
Succeed 137-4
Sum 22-1
Summarize 87-3
Summarize 90-3
Summarize 93-1
Sundry 193-2
Sundry 230-1
Sushi 180-4
Sushi 222-4
Swap 117-1
Tail 115-2
Take 83-2
Tangent 26-3
Times (Complex) 17-4
Times (Complex) 22-4
Times (Complex) 32-4
Times (Polar) 18-2

Times (Polar) 23-2
Times (Polar) 33-2
Tin 180-1
Tin 220-1
Titan 204-4
Titanothere 214-4
Tofu 185-3
Tofu 227-2
Tomb 193-1
Tomb 235-1
Tool 173-4
Transpose 78-4
Trapdoor 155-2
Trapdoor 158-1
Trapper 214-2
Triangle 156-3
Triple 120-1
Truth 7-3
Tuck 119-2
TunaSashimi 182-1
TunaSashimi 223-3
TurtleSashimi 182-4
TurtleSashimi 224-2
U Factor 67-2
UnApprox 13-4
Unbox 87-2
Unbox 89-2
Unbox 90-2
Unbox 95-2
Unequal 12-2
Uni 183-2
Uni 224-4
Union 98-2
Union 100-2
Union 104-3
Up 163-2
Vampire 204-3
VampireBat 204-1
VampireLady 205-2
VampireLeader 205-1
VampireLord 205-3
VirtualDistance 169-1
VirtualDistance 239-2
VirtualHorizontal 169-3
VirtualHorizontal 238-4
VirtualSpace 155-3
VirtualVertical 169-4
VirtualVertical 239-3
Wand 172-2
WaterElemental 199-4
WaterNymph 218-1
Weapon 172-4
WhaleSashimi 181-2
WhaleSashimi 222-3
Where 100-4
Where 162-1
WizardOfYendor 219-1
WoodElf 210-2
WoodNymph 217-4
Wumpus 214-1
Wuxia 7-1
Wye 72-4
Zee 72-2
Zero 112-4
Zip 90-4

Glyph Index

! (U+0021) 19-1, 29-1, 35-1
% (U+0025) 19-2, 24-1, 34-1
& (U+0026) 58-1
* (U+002A) 24-2, 34-2
+ (U+002B) 17-1, 21-1, 32-1
- (U+002D) 17-2, 21-2, 32-2
/ (U+002F) 107-1, 112-1
< (U+003C) 11-1
= (U+003D) 12-1
> (U+003E) 11-2
? (U+003F) 45-1
@ (U+0040) 162-1, 216-1, 217-1, 217-2, 217-3 217-4, 218-1
\ (U+005C) 108-1, 113-1
~ (U+007E) 9-1
¡ (U+00A1) 125-1
¢ (U+00A2) 163-1, 212-1
£ (U+00A3) 162-2
¤ (U+00A4) 212-2
¥ (U+00A5) 219-1
§ (U+00A7) 162-3, 219-2
¬ (U+00AC) 9-2
® (U+00AE) 125-2
± (U+00B1) 130-1, 212-3
¶ (U+00B6) 162-4
½ (U+00BD) 39-1, 44-1
¿ (U+00BF) 125-3, 214-1
Ø (U+00D8) 213-1
ö (U+00F6) 215-1
÷ (U+00F7) 24-3, 34-3
ø (U+00F8) 187-1, 215-2, 236-1, 239-1
Ĝ (U+011C) 203-1
ğ (U+011E) 203-2
Ġ (U+0120) 202-1
Ģ (U+0122) 202-2
Ī (U+012A) 155-1, 237-1
ƀ (U+0180) 204-1
Ɠ (U+0193) 203-3
Ƹ (U+01B8) 215-3
Ǥ (U+01E4) 204-2
Ǧ (U+01E6) 203-4
Ȣ (U+0222) 215-4
Ƚ (U+023D) 216-2
Λ (U+039B) 128-1
Ψ (U+03A8) 213-2, 213-3
α (U+03B1) 145-1
γ (U+03B3) 40-1
ε (U+03B5) 145-2
π (U+03C0) 39-2
ρ (U+03C1) 79-1, 84-1
ϕ (U+03C6) 40-2
ω (U+03C9) 145-3
ϒ (U+03D2) 213-4
Ϝ (U+03DC) 29-2
Ќ (U+040C) 202-3
Э (U+042D) 216-3
Ѫ (U+046A) 206-1, 206-2, 206-3, 206-4, 207-1 207-2, 207-3, 207-4
Ѝ (U+048A) 214-2
ҝ (U+049B) 202-4
Ҟ (U+049C) 201-1
ҟ (U+049D) 201-2
Ӄ (U+04F4) 201-3
ӄ (U+04F5) 201-4
Ṁ (U+1E40) 214-3
ṽ (U+1E7D) 205-1
ỿ (U+1E7F) 204-3
‰ (U+2030) 41-1
⁝ (U+203D) 168-1
⁇ (U+2047) 205-2, 205-3, 205-4, 211-1, 212-4 218-2, 218-3, 218-4, 219-3, 219-4 237-2, 237-3, 237-4, 238-1, 238-2
Ɛ (U+2107) 40-3
ℊ (U+210A) 210-1
ℓ (U+2113) 57-1
№ (U+2116) 76-1

ℙ	(U+2119)	72-1
ℜ	(U+211C)	60-1
ℝ	(U+211D)	172-1
℞	(U+211E)	57-2
ℤ	(U+2124)	72-2, 208-1, 208-2, 208-3, 208-4 209-1, 209-2, 209-3, 209-4
ℨ	(U+2128)	29-3
ℭ	(U+213A)	214-4
ℾ	(U+213E)	73-1, 79-2, 84-2
ℿ	(U+213F)	73-2, 79-3, 83-1
⅁	(U+2141)	204-4
⅋	(U+214B)	58-2, 60-2
⅏	(U+214F)	216-4
Ⅱ	(U+2161)	113-2
Ⅲ	(U+2162)	120-1
↑	(U+2191)	83-2
↓	(U+2193)	83-3
⇋	(U+21CB)	117-1
⇒	(U+21D2)	13-1
↤	(U+21F7)	119-1
↦	(U+21F8)	119-2
↭	(U+21F9)	119-3
∀	(U+2200)	13-2
∂	(U+2202)	57-3
∃	(U+2203)	14-1
∄	(U+2204)	14-2
∅	(U+2205)	7-2
∇	(U+2207)	125-4
∈	(U+2208)	103-1
ϵ	(U+220A)	101-1, 104-1
∐	(U+2210)	80-1, 103-2
∑	(U+2211)	22-1
∔	(U+2214)	110-1
√	(U+221A)	22-2
∩	(U+2229)	98-1, 100-1, 104-2
∪	(U+222A)	98-2, 100-2, 104-3
∫	(U+222B)	57-4
∷	(U+2237)	110-2
÷	(U+223B)	76-2
≈	(U+2248)	13-3
≉	(U+2249)	13-4
≖	(U+2256)	89-1
≗	(U+2257)	89-2
≠	(U+2260)	12-2
≡	(U+2261)	9-3, 11-3
≣	(U+2263)	11-4
≤	(U+2264)	12-3
≥	(U+2265)	12-4
⋬	(U+226C)	107-2, 112-2
⊂	(U+2282)	87-1, 90-1, 95-1
⊃	(U+2283)	87-2, 90-2, 95-2
⊏	(U+2291)	168-2
⊕	(U+2295)	148-1, 151-1, 199-1
⊖	(U+2296)	149-1, 151-2
⊗	(U+2297)	64-1, 66-1, 69-1
⊘	(U+2298)	149-2, 152-1
⊙	(U+2299)	166-1
⊚	(U+229A)	87-3, 90-3, 93-1
⊛	(U+229B)	149-3, 151-3, 199-2
⊞	(U+229E)	80-2
⊠	(U+22A0)	19-3, 36-1
⊡	(U+22A1)	152-2
⊣	(U+22A3)	172-2
⊤	(U+22A4)	7-3, 200-1
⊢	(U+22A6)	200-2
⊶	(U+22B0)	90-4
⊲	(U+22B2)	148-2
⊳	(U+22B3)	148-3
⊷	(U+22B6)	21-3
⊸	(U+22B7)	110-3
⊹	(U+22B8)	93-2
⊹	(U+22B9)	134-1, 140-1
⋄	(U+22C4)	108-2
⋅	(U+22C5)	64-2, 66-2, 69-2
⋈	(U+22C8)	64-3, 66-3, 69-3
⋔	(U+22D4)	120-2
⋕	(U+22D5)	75-1
⋖	(U+22D6)	134-2, 140-2
⋘	(U+22D8)	148-4, 151-4
⋮	(U+22EE)	137-1, 140-3
⋯	(U+22EF)	138-1
⋲	(U+22F2)	78-1
⋳	(U+22F3)	78-2
⋵	(U+22F5)	103-3

⊆	(U+22F8)	100-3	❁	(U+2741)	173-1
⋹	(U+22F9)	103-4	❌	(U+274C)	134-3
⊏	(U+22FF)	168-3	❎	(U+274E)	134-4
⌈	(U+2308)	25-1, 35-2	❖	(U+2756)	186-2, 227-1
⌊	(U+230A)	25-2, 35-3	❗	(U+2757)	174-2
⌬	(U+232C)	117-2	❶	(U+2776)	189-1, 220-2
⌼	(U+233C)	67-1	❷	(U+2777)	189-2, 220-3
⌽	(U+233D)	78-3, 83-4	❸	(U+2778)	189-3, 220-4
⌿	(U+233F)	107-3, 112-3	❹	(U+2779)	190-1, 221-1
⍀	(U+2340)	108-3, 113-3	❺	(U+277A)	190-2, 221-2
⍉	(U+2349)	78-4	❻	(U+277B)	180-2, 222-1
⍋	(U+234B)	73-3	❼	(U+277C)	180-3, 236-2
⍒	(U+2352)	73-4	❾	(U+277E)	190-3, 221-3
⍗	(U+2357)	30-1	❿	(U+277F)	190-4, 221-4
⍰	(U+235B)	18-1, 23-1, 33-1	⟱	(U+27D2)	51-2
⍠	(U+2360)	127-1	⫤	(U+27DA)	82-1
⍣	(U+2363)	122-1	⟰	(U+27F0)	169-1, 239-2
⍳	(U+2373)	76-3	⟲	(U+27F2)	117-3
⍷	(U+237C)	28-1, 35-4	⟳	(U+27F3)	117-4
⎈	(U+2388)	80-3	⤚	(U+291A)	120-3
⎊	(U+238A)	53-1	⤺	(U+293A)	119-4
⎢	(U+23A2)	50-1, 53-2	⦃	(U+2980)	53-3
⏁	(U+23C1)	51-1	⦑	(U+2991)	115-1
⑩	(U+2469)	42-1	⦒	(U+2992)	115-2
⑳	(U+2473)	186-1, 233-1	⦙	(U+2999)	137-4
Ⓒ	(U+24B8)	166-2	⦵	(U+29B5)	169-2, 238-3
Ⓘ	(U+24BE)	166-3	⦶	(U+29BD)	79-4, 84-3
▲	(U+25B2)	163-2	⦺	(U+29CA)	28-2
▼	(U+25BC)	163-3	⧓	(U+29D3)	169-3, 238-4
●	(U+25CF)	174-1	⧗	(U+29D7)	169-4, 239-3
◝	(U+25DD)	25-3	⧾	(U+29DE)	115-3
◞	(U+25DE)	25-4	⨂	(U+2A02)	18-2, 23-2, 33-2
◶	(U+25F6)	26-1	⨃	(U+2A03)	42-2, 44-2
◷	(U+25F7)	26-2	⨊	(U+2A0A)	75-2, 82-2
☘	(U+2618)	210-2, 210-3, 210-4, 211-2	⨢	(U+2A22)	17-3, 22-3, 32-3
☻	(U+263B)	177-1	⨣	(U+2A23)	76-4
⚁	(U+2681)	41-2	⨦	(U+2A27)	21-4
⚃	(U+2683)	41-3	⨫	(U+2A2C)	18-3, 23-3, 33-3
⚅	(U+2685)	41-4	⨯	(U+2A2F)	19-4, 24-4, 34-4
✔	(U+2714)	137-2	⨰	(U+2A30)	17-4, 22-4, 32-4
✧	(U+2727)	137-3	⨻	(U+2A33)	127-2
✪	(U+272A)	180-1, 220-1	⨷	(U+2A37)	89-3, 93-3, 199-3

⨸	(U+2A38)	18-4, 23-4, 33-4, 199-4
⩅	(U+2A42)	100-4
⩃	(U+2A43)	104-4
⩍	(U+2A4D)	28-3
⩓	(U+2A53)	200-3
⩛	(U+2A55)	200-4
⩞	(U+2A5E)	75-3, 82-3
⩟	(U+2A5F)	29-4
⩩	(U+2A69)	51-3, 54-1
⩸	(U+2A78)	110-4
⪦	(U+2AA6)	27-1
⪧	(U+2AA7)	27-2
⪽	(U+2ABD)	27-3
⪾	(U+2ABE)	28-4
⫍	(U+2ACD)	127-3
⫎	(U+2ACE)	127-4
⫏	(U+2ACF)	108-4, 115-4
⫑	(U+2AD1)	27-4
⫖⊂	(U+2AD7)	26-3
⫗≺	(U+2AD8)	26-4
⫩	(U+2AE9)	93-4
⫮	(U+2AEE)	50-2, 53-4
⫵	(U+2AF5)	89-4
⫷	(U+2AF7)	113-4, 122-2
⫽	(U+2AFD)	60-3
⬄	(U+2B04)	152-3
𞸙	(U+2E19)	211-3
支	(U+2F40)	211-4
〘10〙	(U+3248)	193-1, 235-1
〘20〙	(U+3249)	194-1, 235-2
㉾	(U+3267)	193-2, 230-1
㊀	(U+3280)	181-1, 222-2
㊁	(U+3281)	181-2, 222-3
㊂	(U+3282)	181-3, 223-1
㊃	(U+3283)	181-4, 223-2
㊄	(U+3284)	182-1, 223-3
㊅	(U+3285)	182-2, 223-4
㊆	(U+3286)	182-3, 224-1
㊇	(U+3287)	182-4, 224-2
㊈	(U+3288)	183-1, 224-3
㊉	(U+3289)	183-2, 224-4
㊊	(U+328A)	183-3, 225-1
㊋	(U+328B)	183-4, 225-2
㊌	(U+328C)	194-2, 228-1
㊍	(U+328D)	194-3, 228-2
㊎	(U+328E)	194-4, 228-3
㊏	(U+328F)	195-1, 228-4
㊐	(U+3290)	195-2, 229-1
㊗	(U+3297)	180-4, 222-4
㊟	(U+329F)	196-1, 230-2
㊠	(U+32A0)	196-2, 230-3
㊲	(U+32A2)	196-3, 230-4
㊩	(U+32A9)	195-3, 229-2
㊪	(U+32AA)	195-4, 229-3
㊫	(U+32AB)	196-4, 229-4
丽	(U+4E3D)	187-2, 236-3, 239-4
侠	(U+4FA0)	7-1
ꝋ	(U+A74B)	187-3, 236-4
ꭠ	(U+AB44)	61-1
𝆑	(U+1D21D)	155-2, 158-1
𝒜	(U+1D49C)	75-4, 82-4
𝒞	(U+1D49E)	130-2
𝒫	(U+1D4AB)	50-3
𝒱	(U+1D4E5)	155-3
𝔏	(U+1D50F)	58-3
𝔓	(U+1D513)	50-4
𝔸	(U+1D538)	172-3
𝔽	(U+1D53D)	173-2
𝕀	(U+1D540)	166-4
𝕄	(U+1D544)	177-2
𝕆	(U+1D546)	168-4
𝕊	(U+1D54A)	173-3
𝕋	(U+1D54B)	173-4
𝕎	(U+1D54E)	172-4
𝕏	(U+1D54F)	72-3
𝕐	(U+1D550)	72-4
𝔅	(U+1D56D)	39-3, 44-3, 47-1
𝔇	(U+1D56F)	40-4, 44-4, 47-2
𝔍	(U+1D574)	107-4
𝔏	(U+1D577)	66-4
𝔑	(U+1D57D)	39-4
𝔘	(U+1D580)	67-2
𝔝	(U+1D585)	112-4
𝔣	(U+1D58B)	60-4
𝕆	(U+1D7D8)	155-4

𝟙	(U+1D7D9)	156-1
𝟚	(U+1D7DA)	156-2
𝟛	(U+1D7DB)	156-3
	(U+1F000)	188-1, 234-1
	(U+1F001)	188-2, 234-2
	(U+1F002)	188-3, 234-3
	(U+1F003)	188-4, 235-3
	(U+1F029)	189-4, 235-4
	(U+1F02A)	187-4, 234-4
	(U+1F030)	186-3, 231-1
	(U+1F032)	191-1, 231-2
	(U+1F033)	191-2, 231-3
	(U+1F034)	191-3, 231-4
	(U+1F035)	191-4, 232-1
	(U+1F036)	192-1, 232-2
	(U+1F037)	192-2, 232-3
	(U+1F038)	192-3, 232-4
	(U+1F039)	192-4, 233-2
	(U+1F03A)	193-3, 233-3
	(U+1F03B)	193-4, 233-4
手	(U+1F210)	183-4, 225-2
字	(U+1F211)	184-1, 225-3
双	(U+1F212)	184-2, 225-4
デ	(U+1F213)	184-3, 226-1
二	(U+1F214)	184-4, 226-2
多	(U+1F215)	185-1, 226-3
解	(U+1F216)	185-2, 226-4
天	(U+1F217)	185-3, 227-2
交	(U+1F218)	185-4, 227-3
映	(U+1F219)	186-4, 227-4